

TRANSITOPS

Smart Transport Operations Platform

Product Requirements Document

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1. Executive Summary

TransitOps is a centralized transport operations platform that digitizes the full lifecycle of fleet management: vehicle registry, driver management, dispatching, maintenance, fuel and expense tracking, and analytics, while enforcing hard business rules that prevent unsafe or invalid dispatch decisions. This document extends the baseline problem statement with a set of advanced, real-world differentiator modules that go beyond CRUD-and-workflow coverage: computed driver risk scoring, dispatch recommendation intelligence, predictive maintenance, duty-hour compliance, cost leakage detection, a compliance document vault, and a natural-language operations assistant.

The objective is twofold: (1) fully satisfy every mandatory requirement and business rule in the official problem statement, and (2) demonstrate genuine operational intelligence, the kind of system a real logistics company would actually want to run on, so the submission is differentiated from every other team solving the identical brief.

Aspect	Summary
Problem space	Manual, spreadsheet-driven transport operations: scheduling conflicts, missed maintenance, expired licenses, inaccurate costing, no real-time visibility.
Core users	Fleet Manager, Dispatcher/Driver, Safety Officer, Financial Analyst
Baseline scope	Auth + RBAC, Dashboard/KPIs, Vehicle Registry, Driver Management, Trip Management, Maintenance, Fuel & Expense, Reports & Analytics
Differentiators	Computed Safety Score engine, Dispatch Recommendation Engine, Predictive Maintenance, Duty-Hour Compliance, Cost Leakage Detection, Compliance Document Vault, NL Ops Assistant, Live Ops Map, Customer Tracking Link
Delivery format	Responsive web app; role-based dashboards; real-time status updates

2. Problem Statement and Goals

2.1 Background

Many logistics companies still run operations on spreadsheets and paper logbooks. This creates scheduling conflicts, underutilized vehicles, missed maintenance windows, expired driver licenses, inaccurate expense tracking, and near-zero real-time operational visibility for decision-makers.

2.2 Product Goals

- Digitize the complete transport operations lifecycle in a single system of record.
- Enforce hard business rules automatically so invalid states (double-booked drivers, overloaded vehicles, expired licenses on active trips) are structurally impossible, not just discouraged.
- Give every role a real-time, relevant operational view instead of a single generic dashboard.
- Move from reactive operations (fix problems after they happen) to predictive operations (surface risk before it becomes a failure) using the operational data the platform already collects.
- Provide a conversational interface so non-technical managers can query fleet state in plain language instead of navigating multiple screens.

2.3 Non-Goals (Out of Scope)

- Payroll processing or HR management for drivers/staff.
- Native mobile apps (a mobile-first responsive web experience is in scope; native iOS/Android apps are not).
- Hardware/IoT integration for live GPS telemetry (simulated/mock telemetry is used for scoring and risk features; real GPS hardware integration is a future-phase item).
- Accounting/invoicing/GST compliance - expense tracking feeds analytics only, not a ledger.

3. User Roles and Personas

Role	Responsibilities	Core Need
Fleet Manager	Oversees fleet assets, maintenance, vehicle lifecycle and operational efficiency.	Wants a single pane of glass; wants to be warned before something breaks, not after.
Dispatcher / Driver	Creates trips, assigns vehicles/drivers, monitors active deliveries; drivers execute trips and log fuel/expenses.	Wants fast, low-friction trip creation with the system doing the safety-checking automatically.
Safety Officer	Ensures driver compliance, tracks license validity, monitors safety scores.	Wants compliance status to be provable and auditable, not a manual spreadsheet cross-check.
Financial Analyst	Reviews operational expenses, fuel consumption, maintenance costs, profitability.	Wants to spot cost anomalies before they compound across a billing cycle.

4. Baseline Feature Scope (Problem-Statement Mandatory)

These modules satisfy every requirement and mandatory deliverable in the official TransitOps problem statement. They are foundational; the differentiator layer in Section 5 is built on top of this data.

Module	Coverage
Authentication & RBAC	Email/password login, session handling, role-based access control gating every screen and API route by role.
Dashboard	KPIs: Active Vehicles, Available Vehicles, Vehicles in Maintenance, Active Trips, Pending Trips, Drivers On Duty, Fleet Utilization (%). Filterable by vehicle type, status, region.
Vehicle Registry	Master vehicle list: unique Registration Number, Model, Type, Max Load Capacity, Odometer, Acquisition Cost, Status (Available / On Trip / In Shop / Retired).
Driver Management	Driver profiles: Name, License Number/Category/Expiry, Contact, Safety Score, Status (Available / On Trip / Off Duty / Suspended).
Trip Management	Create trips with source, destination, vehicle, driver, cargo weight, planned distance. Lifecycle: Draft to Dispatched to Completed, or Cancelled.
Maintenance	Maintenance records per vehicle; creating an active record auto-flips vehicle status to In Shop and removes it from the dispatch pool.
Fuel & Expense Management	Fuel logs (liters, cost, date) and other expenses (tolls, maintenance); automatic total operational cost (Fuel + Maintenance) per vehicle.
Reports & Analytics	Fuel Efficiency (Distance/Fuel), Fleet Utilization, Operational Cost, Vehicle ROI = $(\text{Revenue} - (\text{Maintenance} + \text{Fuel})) / \text{Acquisition Cost}$. CSV export mandatory, PDF optional.

4.1 Mandatory Business Rules (Hard Constraints)

- Vehicle registration number must be unique.
- Retired or In Shop vehicles must never appear in dispatch selection.
- Drivers with expired licenses or Suspended status cannot be assigned to trips.
- A driver or vehicle already marked On Trip cannot be assigned to another trip.
- Cargo Weight must not exceed the vehicle's maximum load capacity.
- Dispatching a trip automatically sets both vehicle and driver status to On Trip.
- Completing a trip automatically reverts both vehicle and driver status to Available.
- Cancelling a dispatched trip restores vehicle and driver to Available.
- Creating an active maintenance record automatically sets vehicle status to In Shop.
- Closing maintenance restores vehicle to Available (unless the vehicle is Retired).

5. Advanced Differentiator Features

Every team at this hackathon receives the identical problem statement, so baseline coverage alone cannot differentiate a submission. The features below are deliberately not required by the PS but reflect real operational needs of transport/logistics businesses, and are engineered to reuse the same underlying data (trips, fuel logs, maintenance records) the baseline system already captures, maximizing impact per hour of build time.

5.1 Computed Driver Risk & Safety Score Engine [P0 - Core Differentiator | Effort: Medium]

What it is: Replaces the PS's static, manually-entered Safety Score field with a continuously computed score derived from real operational signals: harsh-event flags, on-time delivery rate, fuel-anomaly association, cancelled/incident trip history, and duty-hour violations.

Why it wins: A manually-typed number is trivial to fake and demonstrates no engineering depth. A computed, auditable, tamper-resistant score is exactly what a real Safety Officer needs and what almost no other team will attempt in an 8-hour build.

Key requirements:

- Weighted formula: On-Time Rate (35%) + Incident-Free Rate (30%) + Duty-Hour Compliance (20%) + Fuel-Anomaly-Free Rate (15%), normalized to a 0-100 scale.
 - Score recalculates automatically after every completed trip, incident log, or duty-hour flag.
 - Full score history/audit trail per driver, viewable by Safety Officer and Fleet Manager.
 - Configurable weighting via Settings, so Safety Officer can tune emphasis per organization policy.
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5.2 Dispatch Recommendation Engine [P0 - Core Differentiator | Effort: Medium]

What it is: Instead of a raw dropdown of available drivers and vehicles, the system ranks the best driver-vehicle pairing for a given trip using a scoring model that weighs safety score, vehicle proximity/idle time, cargo-capacity fit, and fleet-wear balancing.

Why it wins: This is the single highest-leverage feature for a live demo: it visibly makes a smart decision in real time instead of just validating a manual choice, and it directly answers the real-world question dispatchers actually struggle with: who/what should I assign, not just is this assignment valid.

Key requirements:

- Ranks all eligible (rule-valid) driver+vehicle combinations by a composite fitness score.
 - Cargo-capacity fit: penalizes large capacity mismatch (assigning a 2-ton truck to a 50kg parcel) to encourage efficient utilization.
 - Fleet-wear balancing: deprioritizes vehicles/drivers with disproportionately high recent utilization to spread wear evenly.
 - Dispatcher can override the recommendation; overrides are logged for later analysis.
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5.3 Predictive Maintenance Alerts [P0 - Core Differentiator | Effort: Medium]

What it is: Extends the reactive maintenance module into a predictive one: the system forecasts upcoming maintenance needs from odometer trend velocity and historical maintenance interval patterns per vehicle, rather than waiting for someone to manually raise a request after something breaks.

Why it wins: The PS's maintenance module is purely reactive. Predicting the need before failure is the difference between a logbook digitization and an actual operations platform, a distinction judges evaluating real-world applicability will notice.

Key requirements:

- Tracks odometer-reading velocity (km/day) per vehicle from trip completions.
 - Compares current odometer against the vehicle's historical maintenance interval (e.g., service every 10,000km) to forecast a predicted service due date.
 - Auto-generates a Maintenance Due Soon alert on the Dashboard and Notifications when a vehicle is within a configurable threshold (e.g., 500km or 7 days) of its predicted due point.
 - Distinguishes forecast-driven alerts visually from reactive fault-reported maintenance records.
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5.4 Duty-Hour & Fatigue Compliance Tracking [P1 - High Value | Effort: Low-Medium]

What it is: Tracks each driver's consecutive driving hours and trips per rolling 24-hour window, flagging drivers who approach or breach safe duty-hour limits, in line with real-world hours-of-service norms for commercial transport.

Why it wins: This is a genuine regulatory/safety concern in real fleet operations that the PS never mentions at all. Building it signals domain understanding beyond implementing the PDF, a strong signal to judges with any operations/logistics background.

Key requirements:

- Rolling-window duty-hour calculation per driver from trip start/end timestamps.
 - Configurable maximum duty-hour threshold (default 10 hours/24h window, adjustable in Settings).
 - Drivers approaching the threshold are flagged amber; drivers who breach it are blocked from new dispatch until reset, feeding the same eligibility check used in Section 5.2.
 - Fatigue-risk indicator surfaced on the Safety Officer dashboard and Dispatch Recommendation Engine.
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5.5 Cost Leakage & Anomaly Detection [P1 - High Value | Effort: Low]

What it is: Reuses the same statistical-baseline approach as driver risk scoring, applied to cost: flags a vehicle whose cost-per-km has spiked significantly against its own historical baseline or against similar vehicles in its class, surfacing possible fuel pilferage, overbilled maintenance, or inefficient routing.

Why it wins: Financial Analysts in real fleets spend significant time manually hunting for exactly this kind of leakage in spreadsheets. Automating the detection is a low-effort, high-perceived-value feature since it reuses fuel/expense data the baseline system already stores.

Key requirements:

- Computes rolling average cost-per-km per vehicle and flags deviations beyond a configurable standard-deviation threshold.
 - Cross-vehicle comparison within the same vehicle Type/class to catch outliers even without individual history.
 - Anomalies surfaced in Reports & Analytics and pushed to the Notification system for the Financial Analyst role.
 - Each anomaly links directly to the underlying fuel log / maintenance record for one-click investigation.
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5.6 Unified Compliance Document Vault [P1 - High Value | Effort: Low]

What it is: Extends the PS's single license-expiry concept into a full compliance calendar covering every document a real commercial vehicle needs: insurance, pollution certificate (PUC), fitness certificate, and permit renewal, each with expiry tracking and tiered reminders.

Why it wins: The PS only asks for license expiry reminders as a bonus feature; real fleets track 4-5 recurring compliance documents per vehicle/driver. This closes a real operational gap cheaply and signals attention to real-world completeness.

Key requirements:

- Document model per vehicle/driver: type, document number, issue date, expiry date, uploaded file reference.
 - Tiered reminders (30 days / 7 days / expired) feeding the Notification system.
 - Vehicles/drivers with any expired mandatory document are automatically excluded from dispatch eligibility, reusing the same rule engine as license/status checks.
 - Single compliance calendar view for the Safety Officer showing all upcoming expiries across the fleet.
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5.7 Natural-Language Operations Assistant [P0 - Core Differentiator | Effort: Medium-High]

What it is: A conversational query interface where a Fleet Manager or Financial Analyst can type plain-language questions such as which vehicles need maintenance this week, show drivers with expiring licenses, or what was our fuel cost last month, and receive an answer generated live from current platform data.

Why it wins: This is the standout live-demo moment: it is visually memorable, interactive, and directly differentiates the team's AI/engineering capability from every other team's static dashboards. It also has the highest wow-factor-per-second of any feature in this document.

Key requirements:

- Natural-language input mapped to a constrained set of pre-defined, safe data queries (no arbitrary code execution against the database).
 - Role-aware answers: a Driver asking the assistant only sees their own trips/data; a Fleet Manager sees fleet-wide answers.
 - Answers rendered as a short natural-language summary plus a supporting table/chart where relevant.
 - Assistant surfaces a graceful fallback for out-of-scope questions rather than hallucinating.
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5.8 Live Fleet Operations Map [P2 - Nice to Have | Effort: Medium]

What it is: A real-time visual map for the Fleet Manager showing all On Trip vehicles with live/simulated position updates, distinguishing normal trips from those flagged by route-deviation or fatigue alerts.

Why it wins: A live, moving map is an immediately compelling visual centerpiece in any demo and very few teams will have real-time visualization working cleanly within the time constraints, a strong return on a moderate amount of build effort.

Key requirements:

- Simulated/mock GPS position feed per active trip (interpolated between source and destination) if hardware telemetry is unavailable.
 - Real-time push of position + status updates to the Manager dashboard (WebSocket-based).
 - Visual distinction for at-risk trips (fatigue flag, cost anomaly, route deviation) directly on the map.
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5.9 Customer-Facing Shipment Tracking Link [P2 - Nice to Have | Effort: Low]

What it is: A shareable, read-only, no-login link per trip that an external customer/client can open to see live shipment status and ETA, the same pattern used by real logistics platforms.

Why it wins: The PS is entirely internal-facing; this feature demonstrates awareness of the actual business model behind transport operations (the end customer cares about their shipment, not the internal dashboard), for very low additional engineering cost.

Key requirements:

- Unique, unguessable tokenized URL generated per trip at dispatch time.
 - Read-only view: status, ETA, and origin/destination; no sensitive internal data (costs, driver personal info) exposed.
 - Link automatically expires/deactivates once the trip is Completed or Cancelled.
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6. Data Model (Key Entities)

Core entities inherited from the baseline problem statement, extended with fields required to power the differentiator features above.

Entity	Key Fields
Vehicle	Registration Number (unique), Model, Type, Max Load Capacity, Odometer, Acquisition Cost, Status, Predicted Next Service Date, Cost-per-KM Baseline
Driver	Name, License Number/Category/Expiry, Contact, Status, Computed Safety Score, Rolling Duty Hours, Score History
Trip	Source, Destination, Vehicle, Driver, Cargo Weight, Planned Distance, Actual Distance, Status, Dispatch Recommendation Snapshot, Public Tracking Token
MaintenanceRecord	Vehicle, Type (Reactive / Predicted), Description, Cost, Opened/Closed Dates, Status
FuelLog / Expense	Vehicle, Liters, Cost, Date, Category (Fuel / Toll / Other), Anomaly Flag
ComplianceDocument	Owner (Vehicle/Driver), Document Type, Document Number, Issue Date, Expiry Date, File Reference, Status
Notification	Recipient Role/User, Type, Message, Related Entity, Read Status, Created At
AssistantQueryLog	User, Query Text, Resolved Intent, Response Summary, Timestamp (for auditability of NL assistant answers)

7. Non-Functional Requirements

Category	Requirement
Performance	Dashboard KPIs and dispatch recommendations must render in under 1.5s for a fleet of up to 500 vehicles.
Reliability	All status-transition business rules (Section 4.1) must be enforced server-side, never trusted from client input alone.
Security	RBAC enforced at API layer; JWT-based auth; passwords hashed (bcrypt or equivalent); public tracking links use unguessable tokens with no PII exposure.
Auditability	Every automatic status change, score recalculation, and NL assistant query is logged with actor, timestamp, and before/after state.
Responsiveness	Fully responsive layout across desktop and mobile viewports for all core screens.
Extensibility	Scoring weights (Safety Score, Cost Anomaly threshold, Duty-Hour limit) configurable via Settings, not hardcoded.

8. Proposed Technology Stack

Layer	Choice
Frontend	React (Vite), Tailwind/Framer Motion for UI polish, Recharts for analytics, Socket.IO-client for real-time updates

Layer	Choice
Backend	Node.js + Express REST API, Socket.IO for live push events (dispatch status, map, notifications)
Database	PostgreSQL with Prisma ORM for type-safe schema and migrations
Scoring/Intelligence Layer	Rule + weighted-scoring engine (Node service) for Safety Score, Dispatch Recommendation, Cost Anomaly, Predictive Maintenance
NL Assistant	LLM-backed intent parser constrained to a defined query schema against live Postgres data, with role-aware response filtering
Auth & Security	JWT, bcrypt password hashing, Express RBAC middleware guards

9. Success Metrics (Demo and Judging Criteria Alignment)

- 100% coverage of mandatory business rules and mandatory deliverables from the official problem statement.
- All core differentiator features (5.1, 5.2, 5.3, 5.7, plus one of 5.4/5.5/5.6) fully functional and demoable end-to-end.
- Zero invalid-state bugs during live demo (double-dispatch, overloaded cargo, expired-license dispatch, etc.).
- NL Ops Assistant answers at least 5 distinct query categories correctly live, on real seeded data.
- Judges can trace at least one differentiator feature back to a concrete real-world operational pain point during Q&A.;

10. Build Prioritization (Offline Round)

Recommended sequencing to guarantee a fully working baseline before layering intelligence, so the team never risks an incomplete core system in favor of unfinished advanced features.

Phase	Scope
Phase 1 - Foundation	Auth/RBAC, Vehicle Registry, Driver Management, Trip Management with all mandatory business rules, Dashboard KPIs
Phase 2 - Baseline Completion	Maintenance workflow, Fuel & Expense tracking, Reports & Analytics with CSV export
Phase 3 - Core Differentiators	Computed Safety Score Engine (5.1), Dispatch Recommendation Engine (5.2), Predictive Maintenance (5.3)
Phase 4 - Standout Feature	Natural-Language Operations Assistant (5.7)
Phase 5 - Stretch (if time remains)	Duty-Hour Compliance (5.4), Cost Leakage Detection (5.5), Compliance Vault (5.6), Live Map (5.8), Customer Tracking Link (5.9)

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